

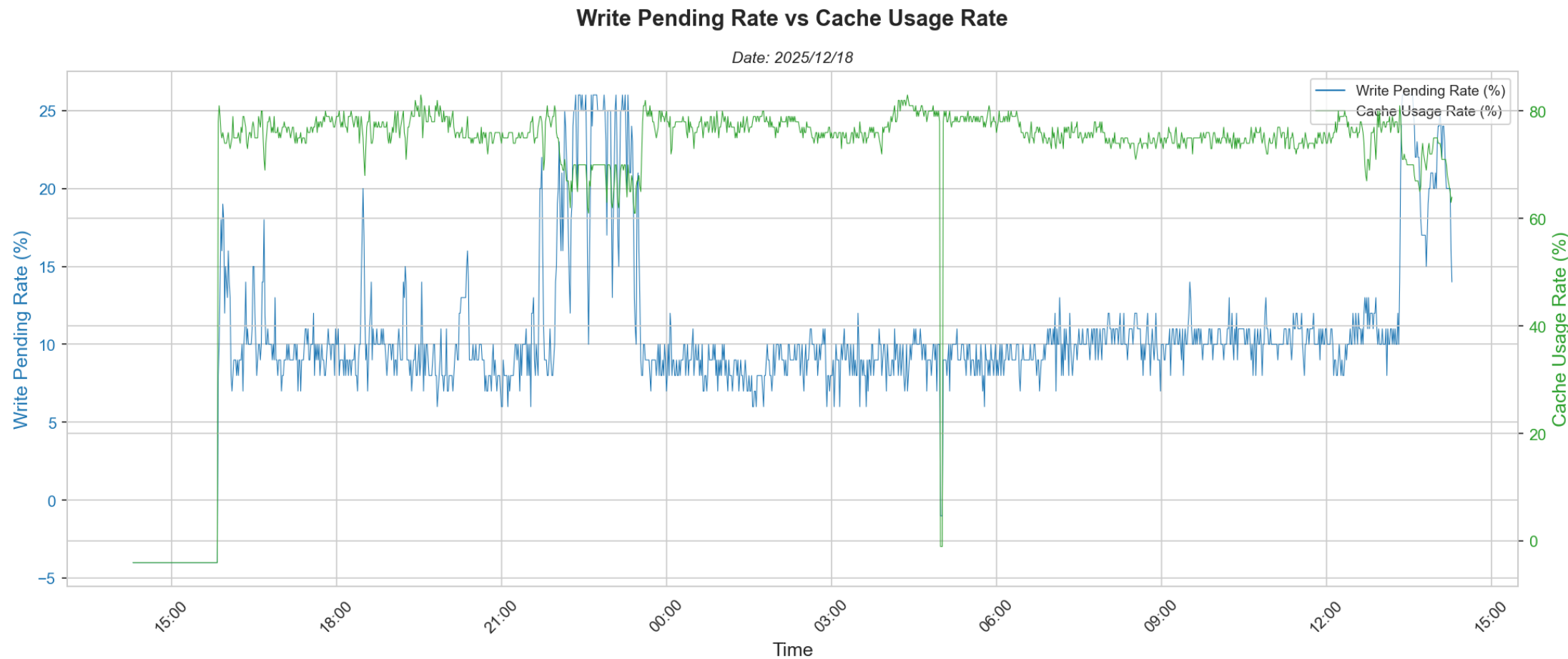
VSP5600 Performance Analysis Report

Storage Performance & Capacity Analysis

Report Generated: 20251230_190225
Data Extraction Date: 20251230_190225

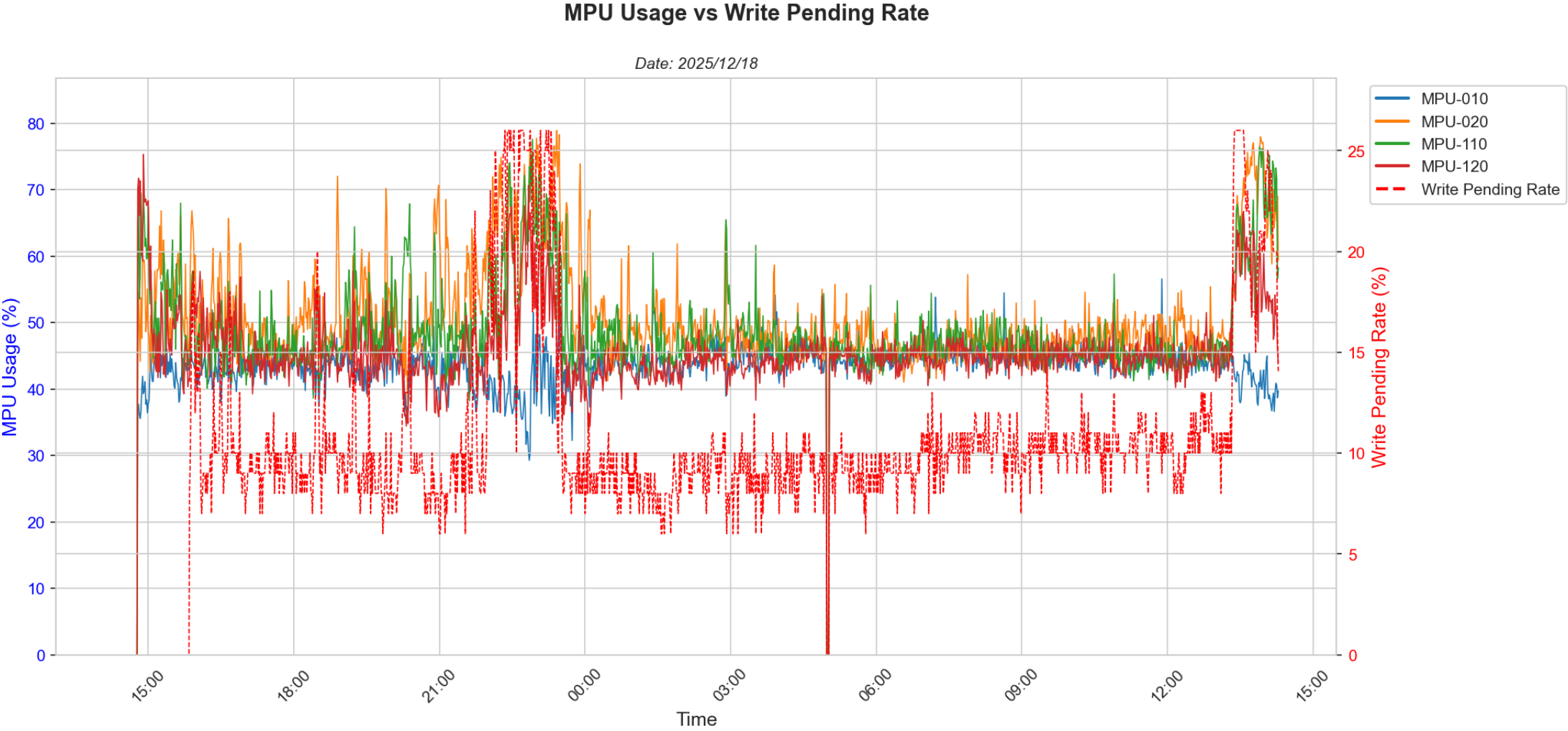
This report provides a comprehensive analysis of storage performance metrics collected from the Hitachi VSP5600 system. It includes detailed charts and analysis of cache performance, processor utilization, port response times, and LDEV (Logical Device) metrics.

6.1 Cache Metrics: Write Pending Rate vs Cache Usage Rate



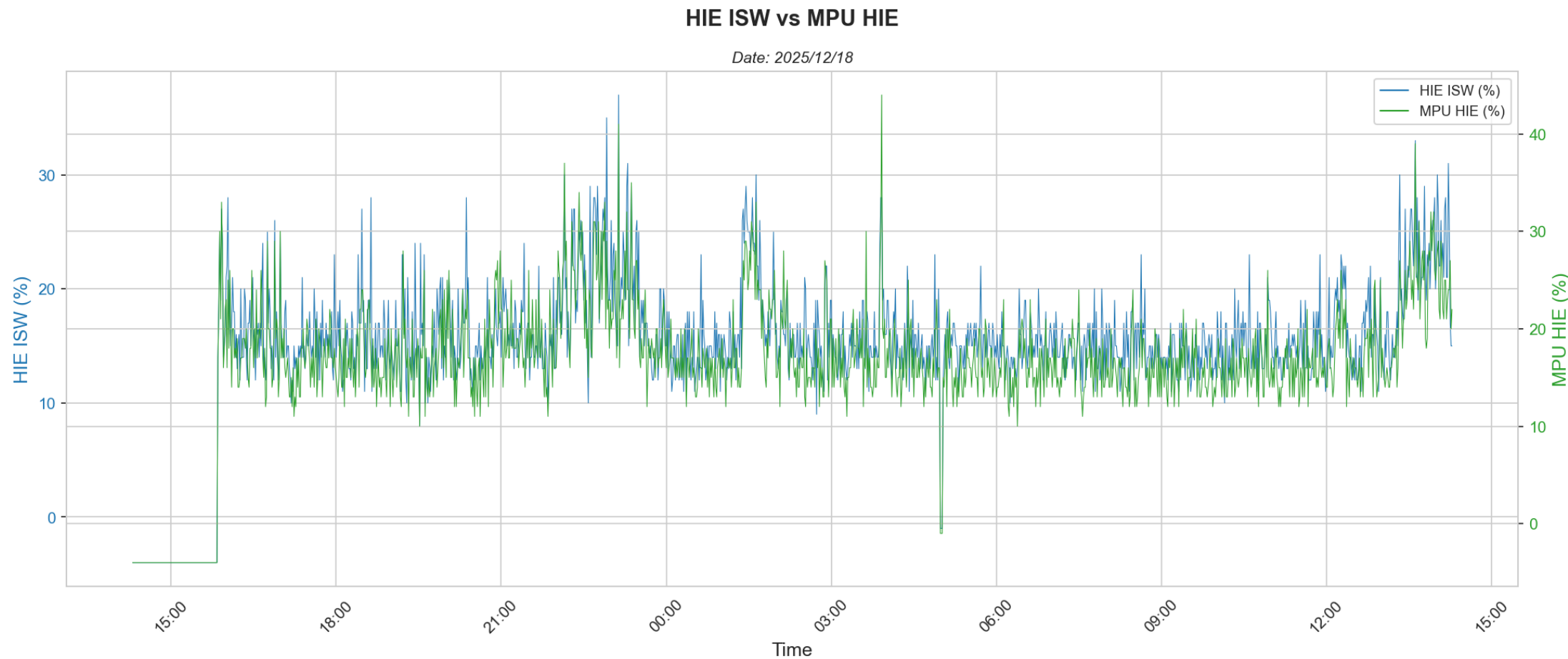
Analysis:
Cache write pending seems to peak to 25% between 21:00 and 00:00 hours. The high write pending indicates a potential issue with cache overflow or poor write performance during peak hours. When write pending increases, it puts pressure on the cache system. If cache usage also increases during this time, the cache memory might be the bottleneck.

6.2 MPU Usage Analysis



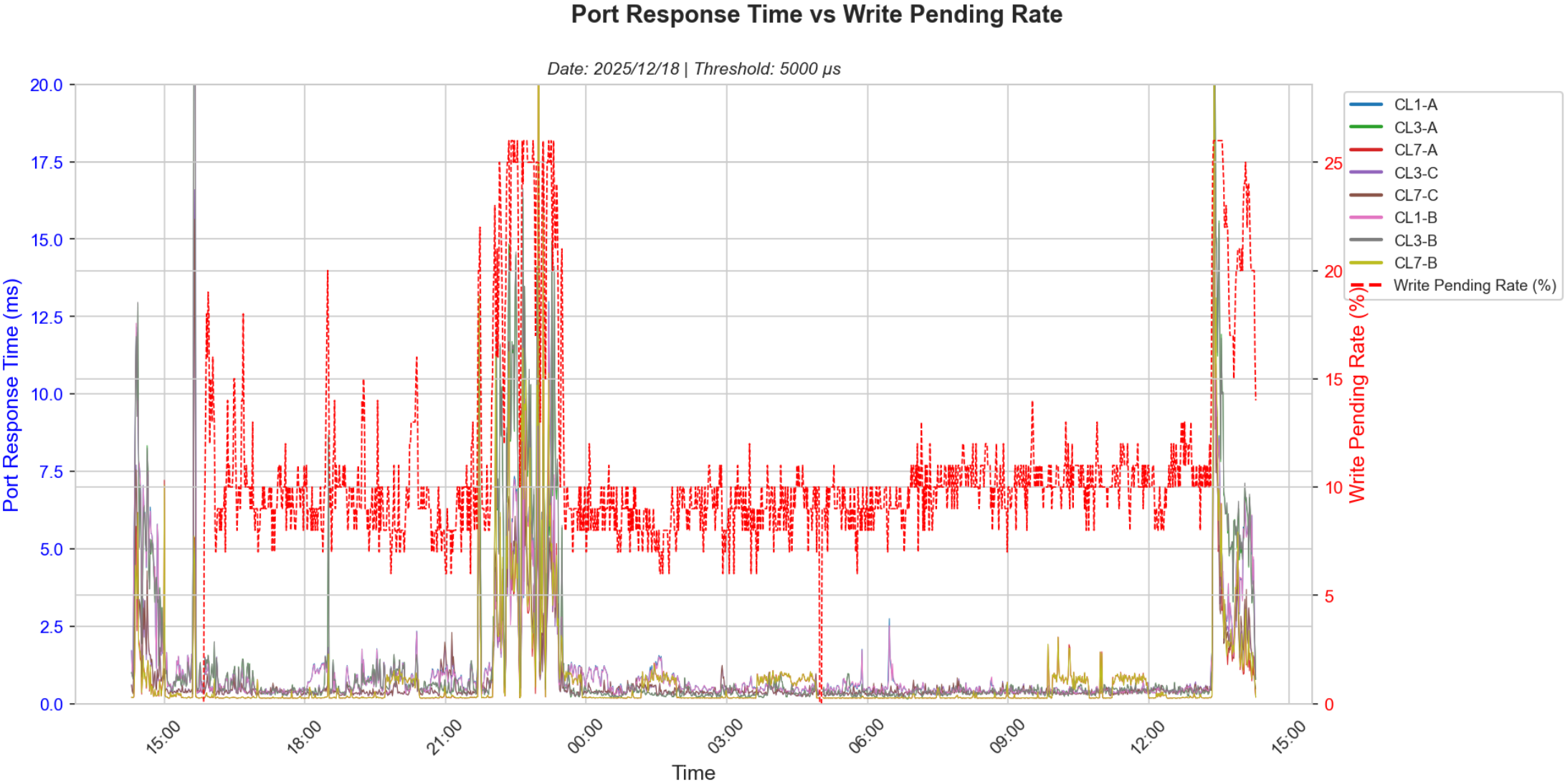
Analysis:
MPU usage is high when write pending is high between 21:00 and 00:00. This correlation indicates that the processor units are heavily utilized during peak write periods. Multiple MPUs show elevated usage, suggesting distributed load across the storage processors, which is normal behavior for a clustered system.

6.3 HIE Metrics: Internal Path Utilization



Analysis:
The chart does not show any saturation of the HIE - ISW and MPU - HIE internal paths. The utilization levels remain well below the saturation point throughout the monitoring period, indicating that the internal communication pathways between processors and the main switch are not a bottleneck for this workload.

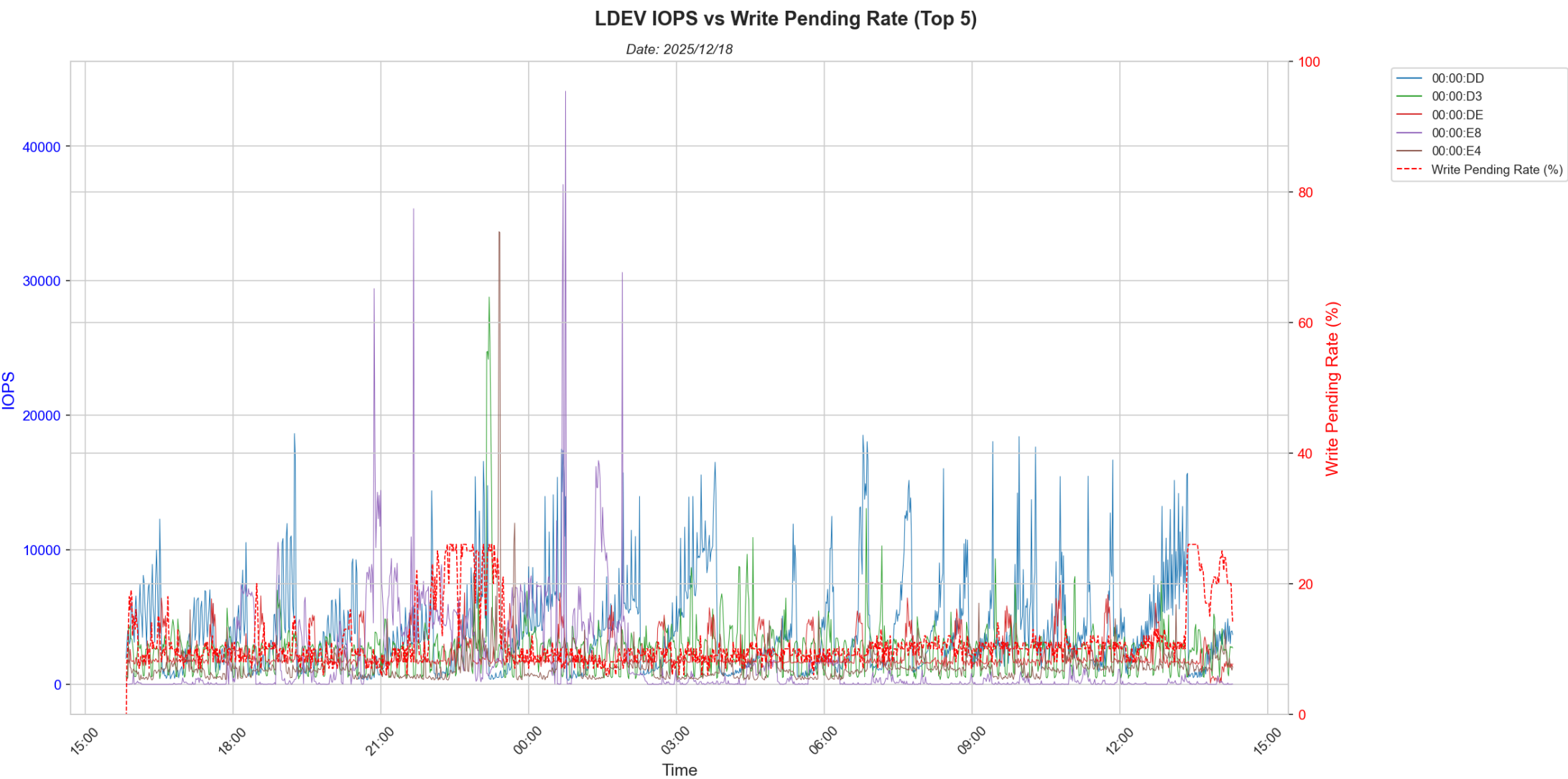
6.4 Port Response Times



Analysis:

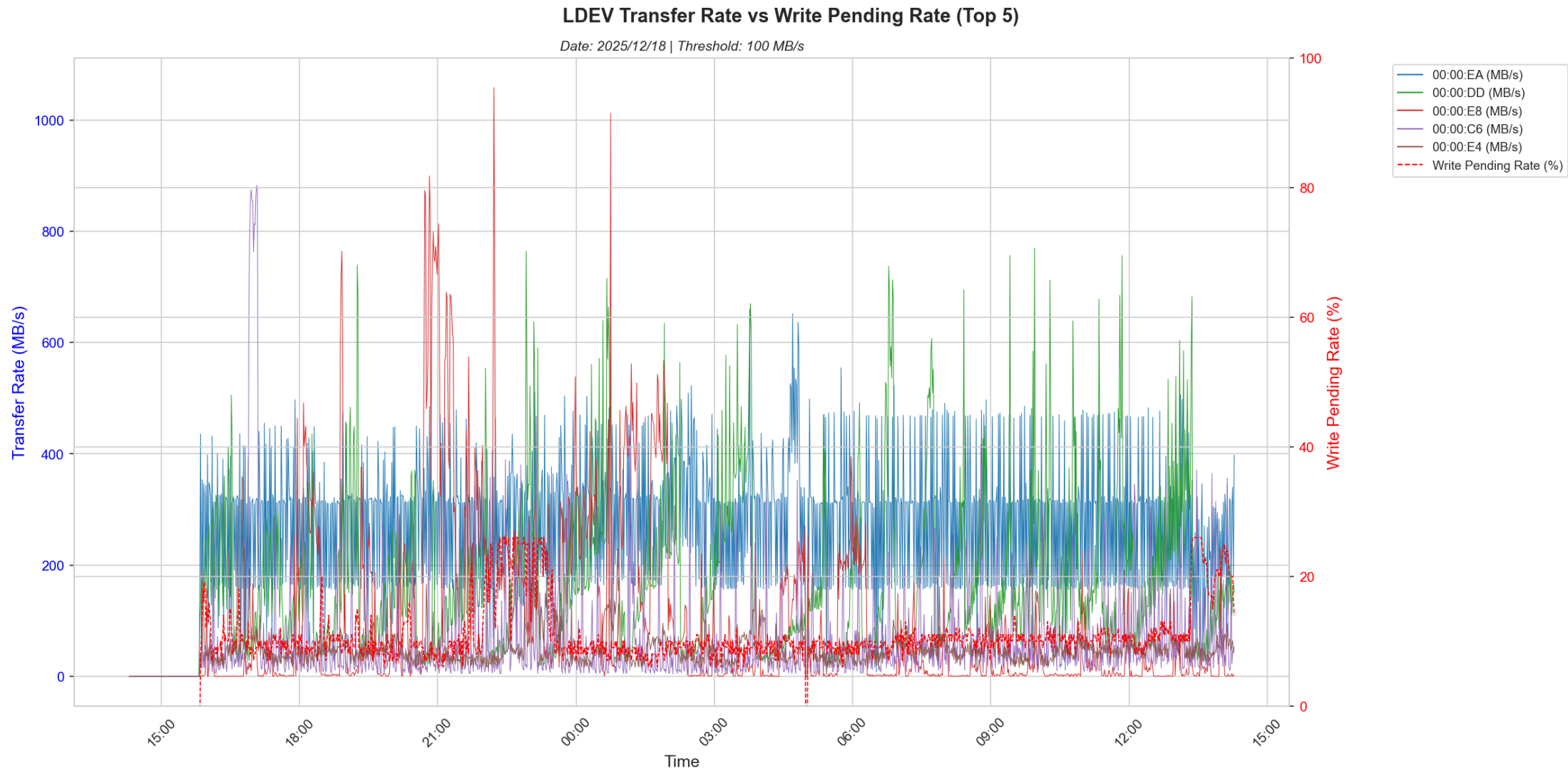
Port response times should be in the range of 1 ms to 3 ms for healthy operation. Response times exceeding this threshold may indicate congestion or performance degradation. The chart shows the correlation between port response times and write pending rate, which helps identify if the delay is related to cache or storage subsystem issues.

6.5 LDEV IOPS Analysis



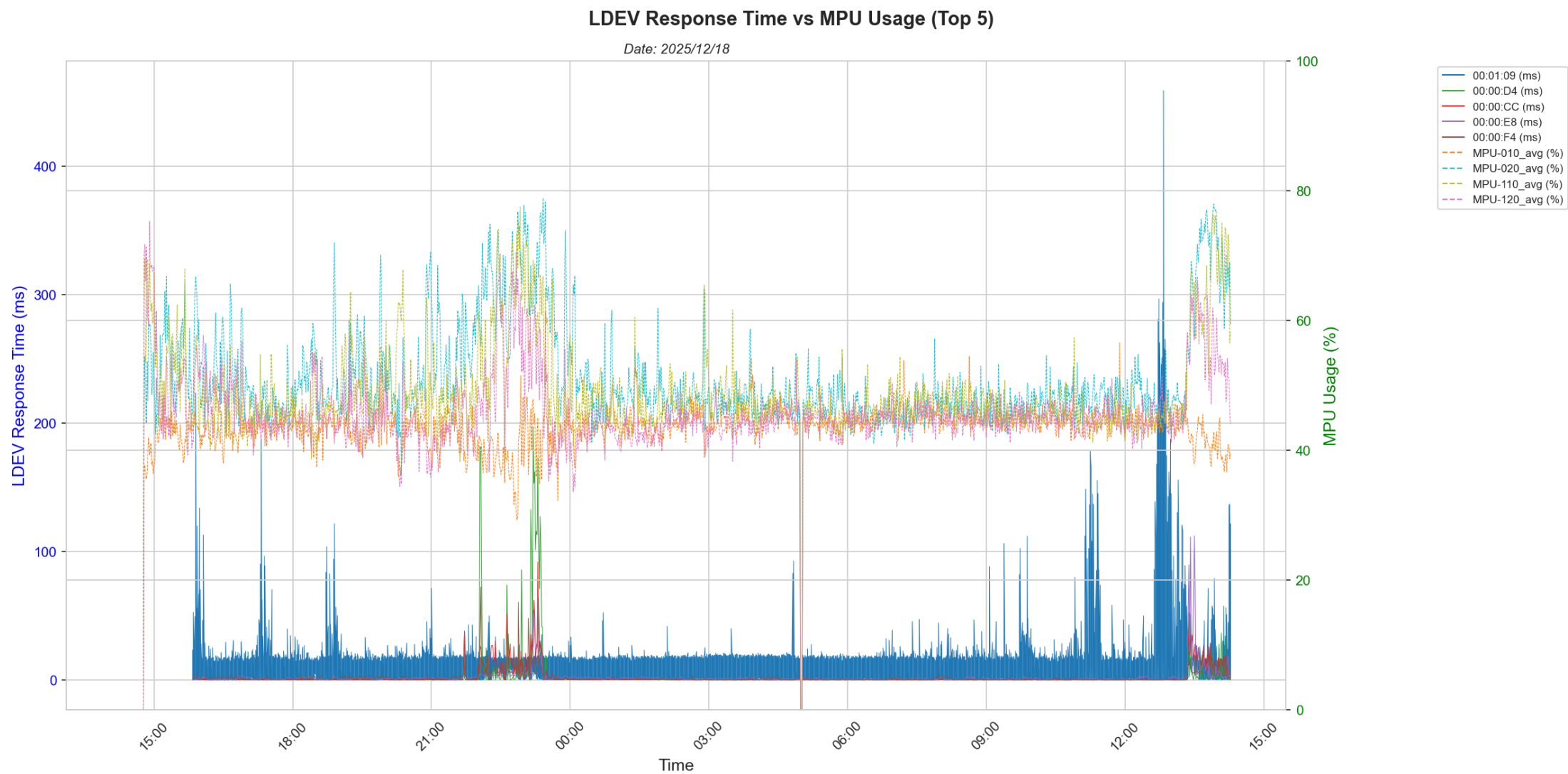
Analysis:
The chart identifies the LDEVs with high IO activity. These are the logical devices that are driving the bulk of the transaction load on the system. High IOPS indicate intensive workload and should be correlated with performance metrics and capacity planning considerations. The top 5 LDEVs account for a significant portion of the total IO activity.

6.6 LDEV Transfer Rate Analysis



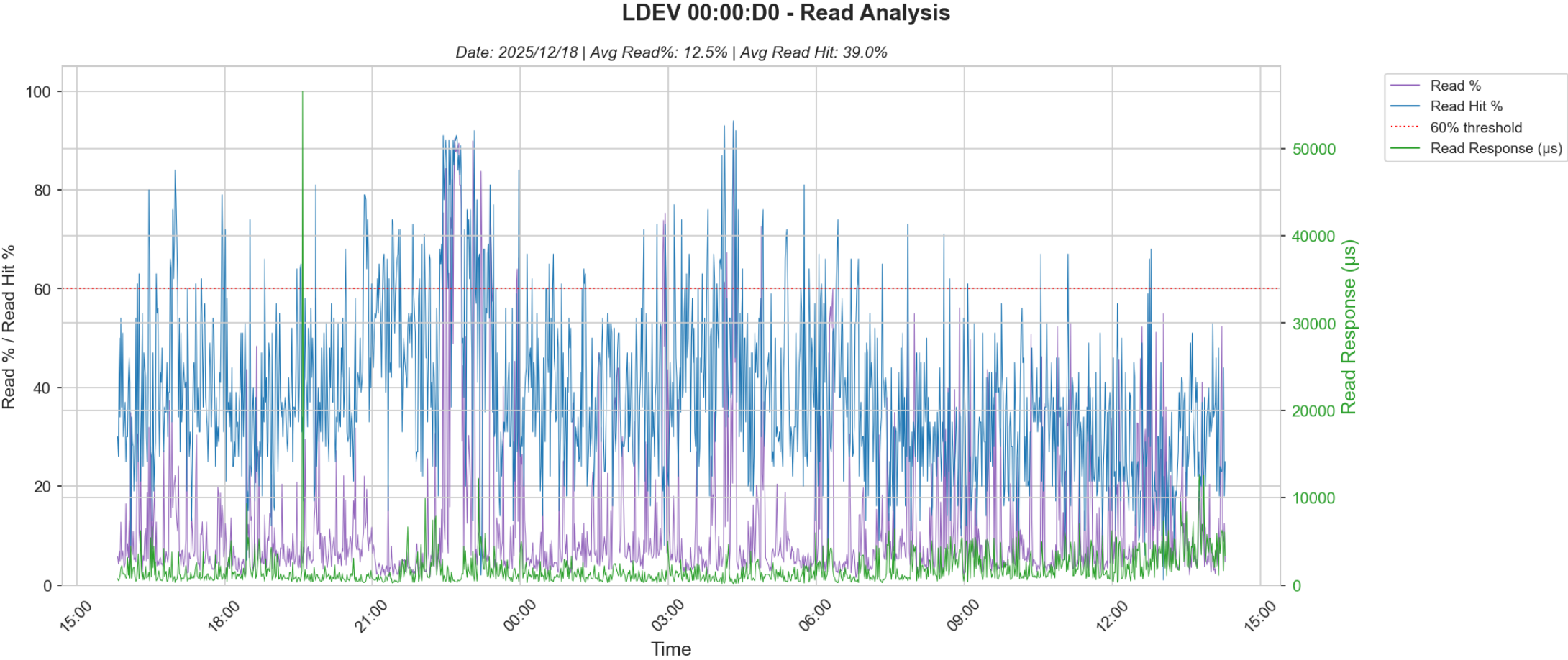
Analysis:
The LDEVs that have high transfer rate are identified in this chart. Sequential data access patterns are indicated by high transfer rates. These LDEVs may be candidates for performance optimization such as RAID level adjustment or dedicated storage tier assignment. The chart shows the throughput patterns throughout the monitoring period.

6.7 LDEV Response Time vs MPU Usage



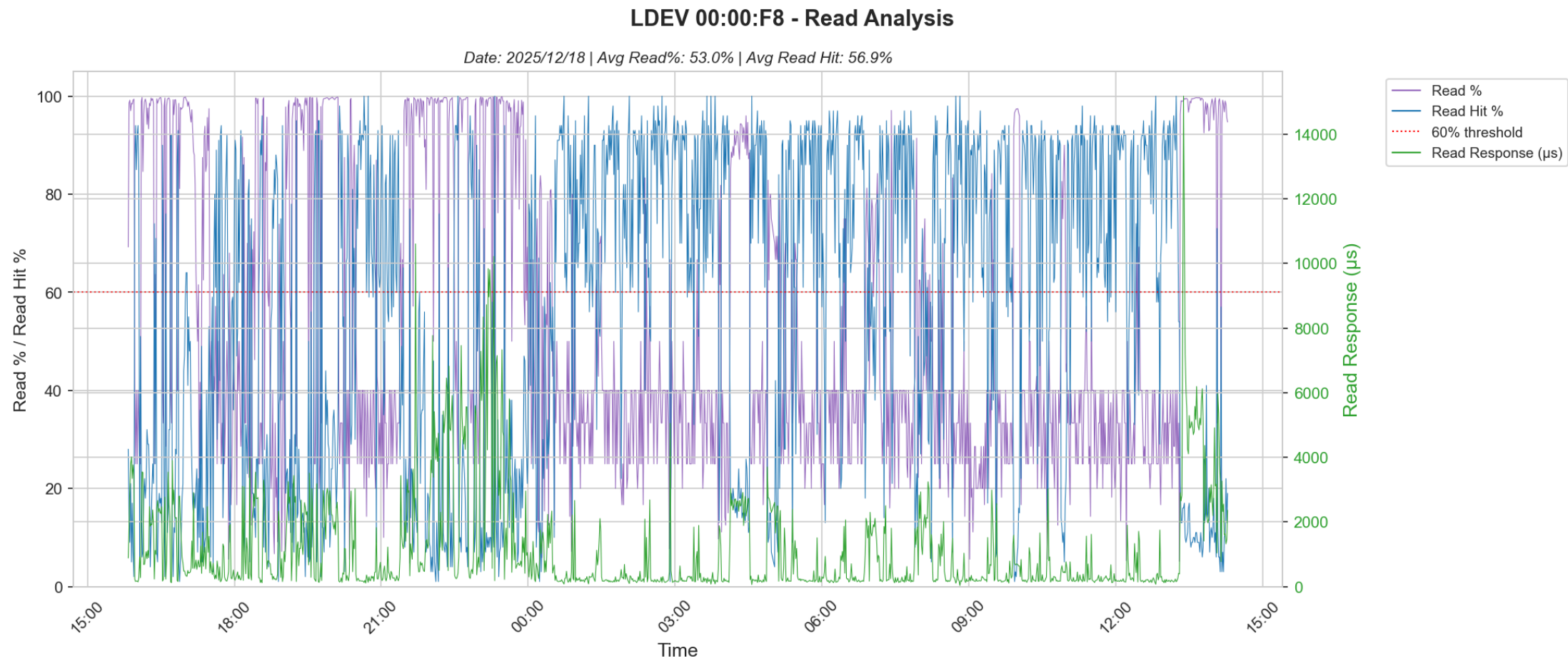
Analysis:
LDEVs that are affected by the high MPU usage are identified when they exhibit increased response times during high MPU utilization periods. This indicates a processor bottleneck affecting IO performance. Response time improvements may be achieved through workload rebalancing or processor upgrade strategies.

6.8 LDEV Read Hit Analysis - 01_00_00_D0X



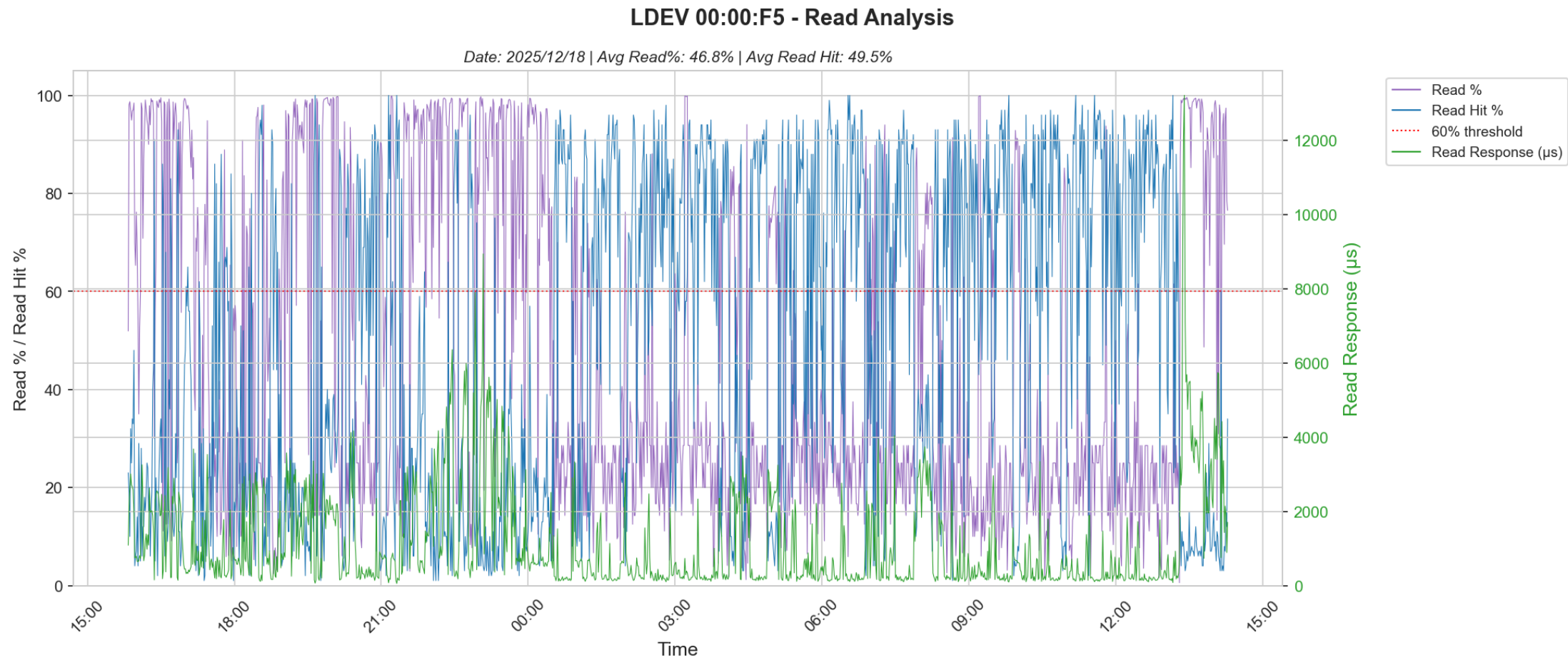
Analysis:
This chart shows the read hit%, read percentage and read response time for read-heavy LDEVs. High read hit percentage indicates good cache effectiveness for read workloads. The correlation between read percentage and read response time helps identify whether the system can efficiently serve read requests or if additional caching or optimization is needed.

6.9 LDEV Read Hit Analysis - 02_00_00_F8X



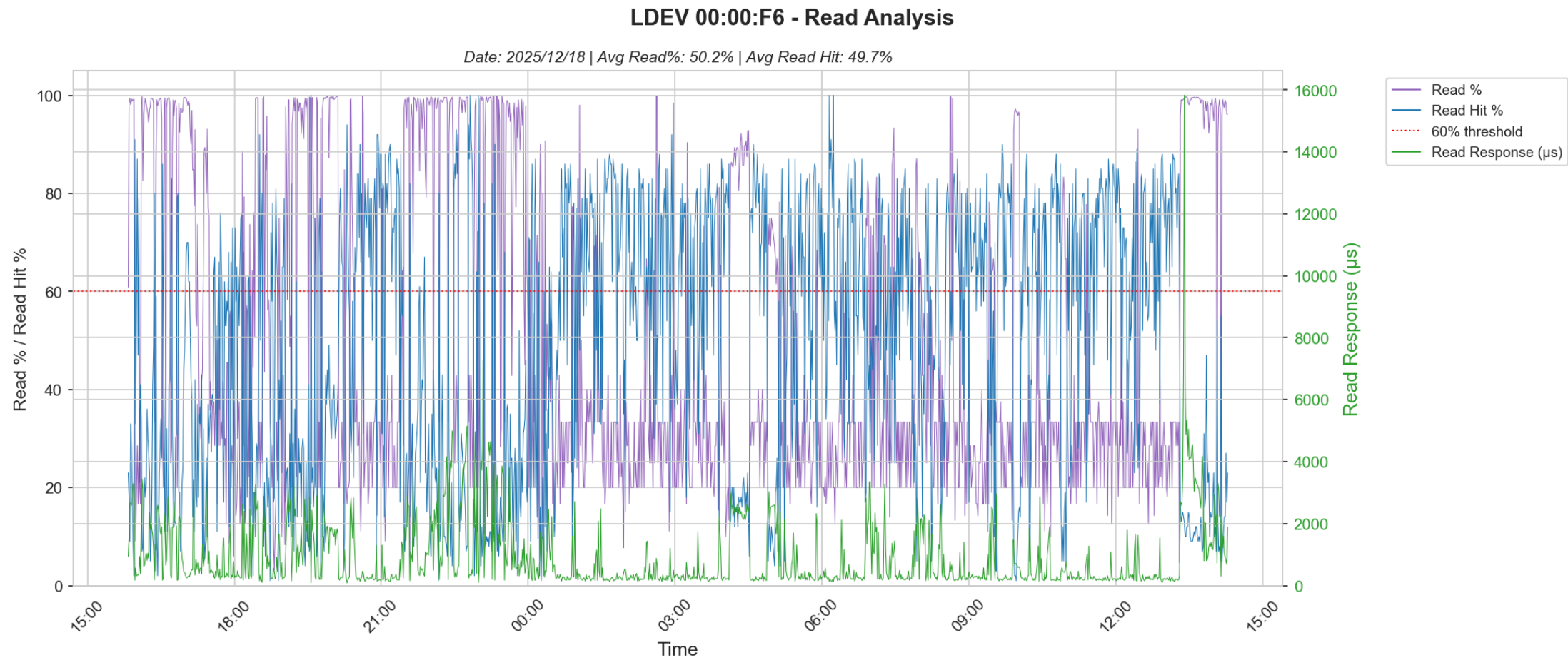
Analysis:
This chart shows the read hit%, read percentage and read response time for read-heavy LDEVs. High read hit percentage indicates good cache effectiveness for read workloads. The correlation between read percentage and read response time helps identify whether the system can efficiently serve read requests or if additional caching or optimization is needed.

6.10 LDEV Read Hit Analysis - 03_00_00_F5X



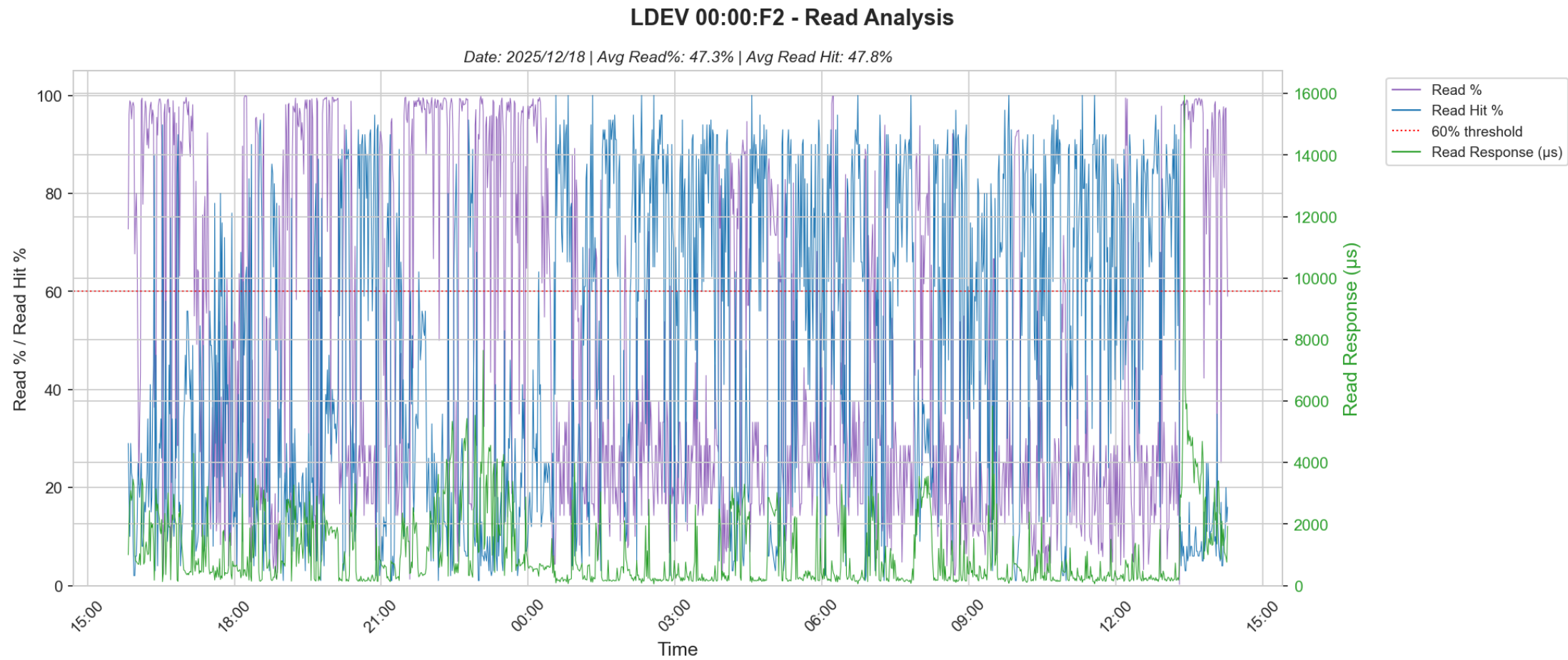
Analysis:
This chart shows the read hit%, read percentage and read response time for read-heavy LDEVs. High read hit percentage indicates good cache effectiveness for read workloads. The correlation between read percentage and read response time helps identify whether the system can efficiently serve read requests or if additional caching or optimization is needed.

6.11 LDEV Read Hit Analysis - 04_00_00_F6X



Analysis:
This chart shows the read hit%, read percentage and read response time for read-heavy LDEVs. High read hit percentage indicates good cache effectiveness for read workloads. The correlation between read percentage and read response time helps identify whether the system can efficiently serve read requests or if additional caching or optimization is needed.

6.12 LDEV Read Hit Analysis - 05_00_00_F2X



Analysis:
This chart shows the read hit%, read percentage and read response time for read-heavy LDEVs. High read hit percentage indicates good cache effectiveness for read workloads. The correlation between read percentage and read response time helps identify whether the system can efficiently serve read requests or if additional caching or optimization is needed.